

PRANAVKUMAR RAJYAGURU

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📍 WITTELSBACHERRING 30, 95444 BAYREUTH | 🇩🇪 NATIONALITY: GERMAN



📁 PROFESSIONAL EXPERIENCE

Project Engineer Structural Simulation & Validation Engineer

From Oct. 2024

Cybex GmbH

- Nonlinear finite-element simulation for structural-integrity assessment under static, dynamic, transient, and highly dynamic loading conditions.
- Development and validation of global and local FE models for structural sizing, stiffness evaluation, and dynamic-response assessment.
- Dynamic and modal analysis to evaluate vibration behaviour and structural response of complex mechanical systems.
- Load-path analysis and assessment of structural stresses, stiffness behaviour, and interface loads.
- Derivation of simulation and test load cases from regulatory, customer, and system-level requirements.
- Cascading of global system-level loads to local component and interface level for structural assessment.
- Hand calculations alongside FE modelling to validate assumptions, sanity-check stiffness and stress results, and guide development direction.
- Investigation of structural failure mechanisms, energy-absorption behaviour, and transient loading events.
- Correlation of finite-element simulations with physical test data to validate model behaviour and improve analysis accuracy.
- Evaluation of temperature-dependent structural behaviour and thermal-expansion effects.

Master Thesis - Uncertainty Quantification of Crash Safety Systems

Mar. - Dec. 23

Porsche AG

- Conducted experimental and simulation-based identification of uncertain input variables in safety-critical systems.
- Applied stochastic modelling and data analysis to characterise system-response behaviour.
- Performed uncertainty quantification of test and simulation results.
- Simulated certification-relevant test cases for systems with uncertain input behaviour.
- Analysed experimental and simulation data using Python and CAE post-processing tools.
- Applied unsupervised learning methods including Principal Component Analysis to identify dominant response patterns.
- Used ANSA, Primer, Python, and LS-DYNA for simulation-data analysis and model evaluation.
- Published related work at Airbag Conference 2024, Fraunhofer Institute.

Werkstudent Functional Design and Simulation

Mar. - Dec. 23

Audi AG

- Developed and validated global and local finite-element models for structural sizing, stiffness evaluation, and dynamic-response assessment.
- Performed hand calculations to support structural sizing and validate analytical assumptions before and alongside FE simulation.
- Cascaded full-vehicle load cases down to local structural interfaces to support stiffness and durability assessment.
- Performed dynamic and modal analyses to evaluate vibration behaviour and structural response of complex vehicle systems.
- Supported investigations of vehicle structural response under vibration test-bench excitation.
- Correlated measurement data with simulation models to improve model fidelity.
- Contributed to the evaluation of static and dynamic stiffness behaviour of vehicle structures and interfaces.
- Used simulation and measurement correlation to support engineering decision-making.

Intern Structural Simulation

Sep. 22 - Feb. 23

Porsche Engineering

- Built and refined finite-element models for dynamic structural and impact simulations.
- Performed explicit FEM simulations for highly dynamic and transient load cases under demanding operating conditions.
- Modelled safety-critical structures and investigated failure mechanisms and energy-absorption behaviour.
- Supported the setup of calculation models in ANSA for structural and impact simulation.
- Supported LS-DYNA finite-element calculations and post-processing of simulation results.
- Prepared simulation results and engineering findings for technical presentations.
- Supported adaptation of automated post-processing scripts using HyperGraph and Animator.

🎓 EDUCATION

M.Eng. Aeronautical Engineering

2020-2024

Technische Hochschule Ingolstadt

- Focus: Structural Design & Simulation.
- Master Thesis: Uncertainty Quantification of Safety Systems, Porsche AG.
- Publication: Accounting for and dealing with uncertainty inherent to ATDs in the design of occupant safety systems, Airbag Conference 2024 - Fraunhofer Institute.
- Student Engineering Team: Formula Student - Structural Design & Aerodynamics.
- Student Engineering Team: Swissloop Tunneling (ETH Zürich) - Structural Design.

B.Eng. Mechanical Engineering
Gujarat Technological University, India

2013-2017

PROJECTS

Development of Fixed-Wing UAV

From Feb. 2025

- Performed iterative structural modifications following flight-test observations and landing loads.
- Balanced structural stiffness, mass and manufacturability throughout UAV development.
- Integrated structural and flight-system design to improve overall aircraft robustness.
- Defined system architecture including IMU, GPS, airspeed sensor, flight controller, and telemetry systems.
- Developed and tuned control loops (PID) to achieve stable and predictable flight behavior under real-world disturbances.
- Conducted iterative flight testing following a structured development cycle (design → build → fly → refine).
- Integrated onboard electronics including ESCs, servos, RC systems, and power distribution for reliable actuation and communication.
- Analyzed flight data and system response to improve robustness against sensor noise, latency, and environmental effects such as wind.

Aircraft Structural and Systems Design

Feb. 2021 - Aug. 2021

- Development of an integrated aircraft simulation model incorporating aerodynamics, propulsion, and environmental conditions.
- Preliminary structural design of wing, empennage, and landing gear.
- Analysis of aircraft dynamic behaviour using flight mechanics simulation.
- Design of flight control and autopilot systems in MATLAB/Simulink.
- Mass and thrust sizing based on mission and performance requirements.

CERTIFICATES

- 2026 - Drone Licence A1/A2/A3 (LBA).
- 2024 - Aerospace Structures and Materials (TU Delft).
- 2024 - Introduction to Model-Based Systems Engineering (Siemens).
- 2024 - Applying Systems Engineering to the Design Process (University of Colorado Boulder).
- 2020 - Essentials of Siemens NX CAD Design (Siemens Center of Excellence).

SKILLS & TOOLS

CAE Ansys, LS-DYNA, ANSA, HyperMesh, OptiStruct

CAD CATIA V5/V6

Programming Python, Shell Scripting, Data Analytics, Machine Learning

LANGUAGES & HOBBIES

Languages German: C1 (fluent), English: B2 (upper-intermediate)

Hobbies Drohnen Pilot Kitzrettung, Model Aircraft (LSG Bayreuth) & Glider, Tennis